

Rirekisho (Curriculum Vitae)

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Name: Naziru Halilu
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Education and Work History

Year / Month	Education and Work History
2016 / 09	Enrolled, National Diploma, Agricultural Engineering Technology, Division of Agricultural Colleges, ABU Zaria
2018 / 07	Completed National Diploma, Distinction, CGPA 3.78/4.00
2019 / 09	Enrolled, B.Eng., Agricultural and Bioresources Engineering, Ahmadu Bello University
2024 / 07	Completed B.Eng., First Class Honours, CGPA 4.77/5.00, top 1% of cohort
2021 / 01	Project Assistant, Smart Irrigation and Water Systems, BOMO Basin Project, Nigeria
2022 / 01	Student Research Assistant, Precision Crop Systems, NAERLS, Nigeria (78% crop stress detection accuracy)
2022 / 06	Assistant Farm Manager (concurrent, part time), Yankari Nasarawa Farms, Nigeria
2024 / 01	Project Assistant, Remote Data Lead, Smart Greenhouse Automation, IAR-ABU, Nigeria
2024 / 03	Undergraduate Project Assistantships, Depts. of Agricultural and Bioresources Engineering and Animal Science, ABU
2025 / 09	Enrolled, Erasmus Mundus Joint Master's Degree (EMJMD), Sustainable Agriculture, UPNA Spain / UTAD Portugal / AUA Greece
2025 / 09	EMJMD Research Scholar, UPNA, Spain. Term 1: technological bases for agricultural sustainability. Built STADOS (39% NDVI variance cut), N_GACL (33.8% accuracy), Digital Twin (47.8% soil loss cut), GIS-Slurry (RMSE 7.78)
2026 / 02	EMJMD Research Scholar, UTAD, Portugal. Term 2: precision agriculture. Built Operationalizing GIS/ML (R sq 0.978-0.995), MSRK (20% uncertainty cut), Brix vineyard forecasting (R sq 0.9784), Grape Moth monitoring
2026 / 06	Summer Intern, Computer Vision Group, ABU. Extended GACL, 64.4% accuracy across 19 disease classes; first author paper under review at Biosystems Engineering
2026 / 09	EMJMD Research Scholar, AUA, Greece (in progress). Term 3: smart irrigation systems. Built NaCROP irrigation DSS
2027 / 06	Expected completion, EMJMD, Sustainable Agriculture (expected)

Licenses and Qualifications

2023: PTDF Undergraduate Fellow 2023: Pupil Engineer, Council for the Regulation of Engineering in Nigeria (COREN) 2025: Erasmus Mundus Joint Master Scholarship, European Union

Research Publications

- Halilu, N., Sulaiman, J. (2026). Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 13(7), 2951 to 2989.
- Halilu, N., Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 13(7), 2990 to 3020.
- Ismail, K., Halilu, N. (2024). AI Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. Proc. AfCPA, 2(1), 100 to 120.
- Halilu, N. (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 13(7), 2920 to 2950.
- Auwal, M., Yusuf, A., **Naziru, H.** (2020). Physio chemical and bacteriological quality of water for irrigation purpose. Nigerian Research Journal of Engineering and Environmental Sciences, 5(1), 198 to 204.
- Halilu, N., Sulaiman, J. Geometry Agnostic Contrastive Learning (GACL). Biosystems Engineering, Elsevier. Under review.

- Halilu, N., Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. European Journal of Agronomy, Elsevier. Under review.
- Halilu, N. (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (*Ipomoea* spp.). Environmental Monitoring and Assessment, Springer Nature. Under review.
- Halilu, N. (2026). Multitask Singularity Regression Kriging for Spatio Temporal Soil NPK Crop Stress Yield Prediction. Zenodo Preprint.
- Halilu, N. (2026). STADOS: Spectral Temporal Adaptive Dark Object Subtraction. Zenodo.
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time Series Modelling and Control Systems in Precision Irrigation. Zenodo.

Skills

Geospatial AI / GIS: QGIS, ArcGIS, ENVI, Google Earth Engine, Sentinel imagery

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, CNNs

Programming: Python, MATLAB, R, Git/GitHub, Zenodo

Reason for Application / Self Introduction

I am an agricultural engineer specialising in UAV and sensor based high throughput phenotyping, computer vision, and geospatial machine learning. I bring a first class academic record, four peer reviewed publications, and fourteen open source research repositories, and I am seeking opportunities to apply reproducible, field validated machine learning systems to real agricultural problems.

References

Prof. Miguel Ángel Campo Bescós, Public University of Navarre, miguel.campo@unavarra.es

Prof. José Tadeu Marques Aranha, UTAD, Portugal, j_aranha@utad.pt

Dr. Ibrahim Babangida Dalha, Ahmadu Bello University, ibdalha@abu.edu.ng